

Name of Student : Arvind Patil

Class : SE

Division : S2

Roll Number : 79

Batch : S21

**Subject : Fundamentals of Telecommunication
Systems**

Experiment Number : 08

**Experiment Title : Demonstration of Signal Transmission through
Wireless Media (Wi-Fi)**

Date of Performance : 23/02/2026

Date of Submission : 23/02/2026

Experiment No. 08: Demonstration of Signal Transmission through Wireless Media (Wi-Fi)

Theory

1. Wireless Media

Wireless media refers to communication channels that transmit data **without using physical cables or wires**. It uses **electromagnetic waves** such as radio waves, microwaves, or infrared signals.

Examples:

- Wi-Fi
- Bluetooth
- Satellite communication

Features:

- No physical connection required
- Easy to install and flexible
- Supports mobility

Conclusion:

Wireless media enables communication over distances without the need for physical transmission cables.

2. SSID (Service Set Identifier)

SSID (Service Set Identifier) is the **name of a wireless network (Wi-Fi network)**.

Purpose:

- Identifies a specific wireless network

- Allows devices to connect to the correct Wi-Fi network
- Helps distinguish between multiple nearby networks

Example:

- “Home_WiFi”
- “Office_Network”

Conclusion:

SSID acts as the **identity or name of a Wi-Fi network** that users select to connect.

3. Difference between Wired LAN and Wireless LAN (WLAN)

Feature	Wired LAN	Wireless LAN (WLAN)
Medium	Uses cables (Ethernet)	Uses wireless signals (Wi-Fi)
Mobility	Limited (fixed connection)	High mobility
Speed	Generally faster and stable	Slightly slower than wired
Installation	Requires cabling	Easy to install
Security	More secure	Less secure (can be hacked)

Conclusion:

- Wired LAN is **faster and more secure**
- WLAN is **more flexible and convenient**

4. Significance of Signal Strength in Wireless Transmission

Signal strength refers to the **quality and power of a wireless signal** between devices.

Importance:

- Strong signal → Faster data transfer and stable connection
- Weak signal → Slow speed, interruptions, or disconnection

Factors affecting signal strength:

- Distance from router
- Physical obstacles (walls, buildings)
- Interference from other devices

Conclusion:

Signal strength is important because it directly affects the **performance and reliability of wireless communication**.

Hardware Components Required

- **Wireless Router (WRT300N):** The central device that broadcasts the wireless signal and manages the network.
- **Laptop-PT (x2):** End devices equipped with wireless network interface cards.
- **Smartphone (x1):** A mobile device with built-in Wi-Fi capability.

Hardware and Connection Setup

- **Wireless Standard:** 2.4 GHz 802.11n
- **Network ID (DHCP):** 192.168.0.0 (Default for WRT300N)
- **SSID (Network Name):** MyWirelessNetwork
- **Authentication:** WPA2-Personal

Procedure

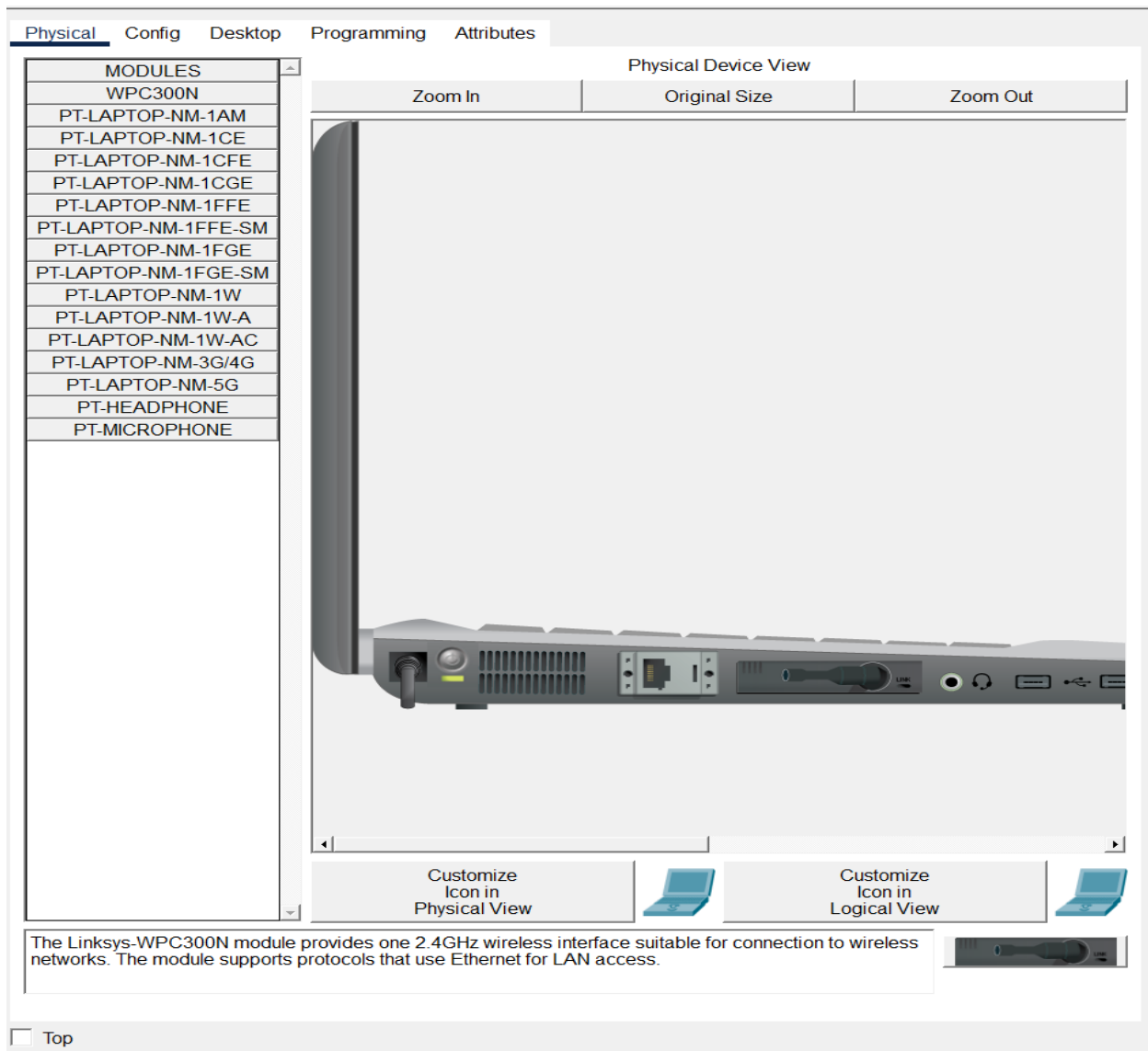
Step 1: Physical Topology Design

- Place one **Wireless Router (WRT300N)** in the center.
- Place two **Laptops** and one **Smartphone** near the router.

- Naming: Rename Smartphone following the convention (e.g., S11_01).
- **Observation:** Note that the Smartphone connects automatically (default settings), but the Laptops do not.

Step 2: Hardware Identification (Laptop Modification)

- Click on a **Laptop** > **Physical** tab.
- **Task:** Turn off the Laptop power. Remove the Ethernet module (RJ45) and drag the **WPC300N** (Wireless Module) into the slot. Turn the power back on.
- **Observation:** A dashed line appears between the Laptop and Router, indicating a wireless link.



Step 3: Wireless Router Configuration

- Click the **Wireless Router > Config tab > Wireless**.
- Set the **SSID** to MyWirelessNetwork.
- Select **WPA2-PSK** and enter a Passphrase (e.g., TSEC@2026).
- **Observation:** The dashed lines disappear because the clients now need the correct security key to connect.

Step 4: Connecting Clients to the Wireless Signal

- Click **Laptop > Desktop tab > PC Wireless**.

- Go to the **Connect** tab. Click **Refresh**. Select MyWirelessNetwork and click **Connect**.
- Enter the Passphrase TSEC@2026
- *Repeat this for the second Laptop and the Smartphone (under Config > Wireless).*

Step 5: Logical Addressing (DHCP Verification)

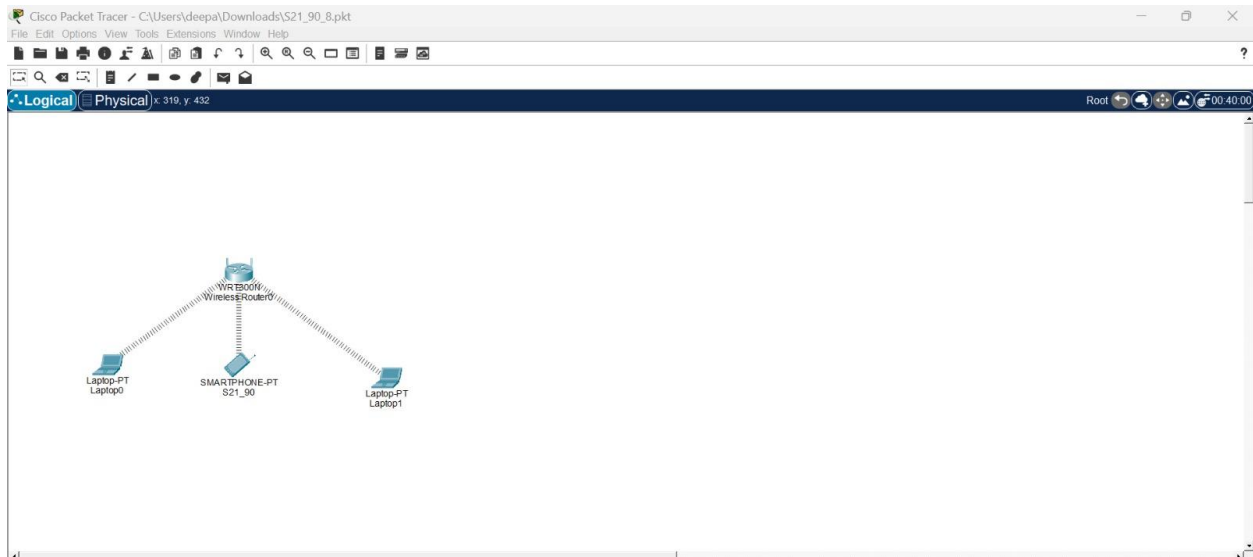
- Unlike the wired experiment, we will not set Static IPs.
- Click **Laptop > Desktop > IP Configuration**. Ensure **DHCP** is selected.
- **Task: Complete the following table with the IPs assigned automatically by the Wireless Router.**

Device	SSID	Authentication Type	Assigned IP Address
Laptop A	MyWirelessNetwork	WPA2-PSK	192.168.0.102
Laptop B	MyWirelessNetwork	WPA2-PSK	192.168.0.103
Smartphone	MyWirelessNetwork	WPA2-PSK	192.168.0.100

Step 6: Testing Signal Transmission

- **Real-time:**
 - Open the Command Prompt on Laptop A and ping the IP address of the Smartphone.
- **Simulation Mode:**
 - Filter for **ICMP**.
 - Use the Add Simple PDU (Envelope) to click Laptop A then Smartphone.

- Watch the packet travel through the air (wireless media) to the Router and then to the target device.



Observations

- **No Physical Cables:** Communication was established without the use of Copper or Fiber cables, proving the use of Radio Frequency (RF) as the transmission media.
- **SSID Filtering:** Devices could only see and connect to the specific network identified by the SSID.
- **DHCP Automation:** The Wireless Router automatically managed the IP addresses, simplifying the setup of a WLAN.

Conclusion

The experiment successfully demonstrates signal transmission through **Wireless Media**. It proves that a **Wireless Router** acts as an Access Point and a DHCP server, allowing mobile and portable devices to communicate within a defined range using radio waves and security protocols.